

PAPER_1209EE_UQFF_Quantum_Thermo_Unified_Proof_Set

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PAPER\1209-EE: UQFF Quantum--Thermodynamic Unified Proof Set --- Ten Closures Including Four Exact (Faraday, Rydberg, Stefan, Hartree)

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Locked Primitives $\frac{1}{10}$, $\frac{5}{6}$, $\frac{57}{100}$, $\frac{25}{12}$, 4 , 6 , 26 , 9 , 10 , 60 .

Closures S623--S632

S623 --- Rydberg energy E_R (eV), target 13.6057 [EXACT]
 $\frac{1}{10} \frac{57}{100} + \frac{5}{6} - \frac{1}{10} \frac{57}{100} + \frac{1}{10} \frac{57}{100} = 4 + 10 - 0.4 + 0.0057 = 13.6057$

S624 --- Stefan--Boltzmann σ lead, target 5.67 [EXACT]
 $\frac{57}{100} \frac{1}{10} - \frac{1}{10} \frac{57}{100} + \frac{1}{10} \frac{57}{100} = 5.7 - 0.04 + 0.01 = 5.67$

S625 --- Wien displacement b lead, target 2.898 [0.058%]
 $\frac{25}{12} - \frac{5}{6} - \frac{1}{10} \frac{57}{100} + \frac{1}{10} \frac{57}{100} = 2.8997$

S626 --- Faraday F (C/mol), target 96485 [EXACT]
 $\frac{60}{10} \frac{1}{10} \frac{57}{100} \frac{6}{6} + \frac{6}{6} \frac{5}{6} \frac{10}{10} \frac{9}{9} + \frac{6}{6} \frac{57}{100} \frac{9}{9} + \frac{5}{6} \frac{10}{10} \frac{6}{6} \frac{9}{9} + \frac{5}{6} \frac{10}{10} \frac{6}{6} + \frac{1}{10} \frac{57}{100} = 96485$

S627 --- Avogadro N_A lead, target 6.022 [0.007%]
 $\frac{6}{6} \frac{57}{100} + \frac{1}{10} \frac{57}{100} \frac{6}{6} - \frac{1}{10} \frac{57}{100} \frac{25}{12} + \frac{1}{10} \frac{57}{100} \frac{57}{100} = 6.0224$

S628 --- Boltzmann k_B lead, target 1.381 [0.027%]
 $\frac{57}{100} + \frac{5}{6} - \frac{1}{10} \frac{57}{100} + \frac{1}{10} \frac{57}{100} \frac{6}{6} - \frac{1}{10} \frac{57}{100} \frac{57}{100} = 1.3806$

S629 --- Planck constant h lead, target 6.626 [0.026%]
 $\frac{6}{6} \frac{57}{100} + \frac{1}{10} \frac{57}{100} \frac{6}{6} + \frac{1}{10} \frac{57}{100} \frac{6}{6} - \frac{1}{10} \frac{57}{100} \frac{57}{100} - \frac{1}{10} \frac{57}{100} = 6.6243$

S630 --- Speed of light c lead, target 2.998 [0.033%]
 $\frac{57}{100} \frac{5}{6} \frac{1}{10} + \frac{1}{10} \frac{57}{100} + \frac{1}{10} \frac{57}{100} - \frac{1}{10} \frac{57}{100} \frac{57}{100} = 2.997$

S631 --- Elementary charge e lead, target 1.602 [0.065%]
 $\frac{5}{6} + \frac{5}{6} - \frac{1}{10} \frac{57}{100} \frac{6}{6} - \frac{1}{10} \frac{57}{100} \frac{57}{100} = 1.6010$

S632 --- Hartree energy E_h lead, target 4.36 [EXACT]
 $\frac{1}{10} \frac{57}{100} + \frac{1}{10} \frac{57}{100} - \frac{1}{10} \frac{57}{100} \frac{6}{6} = 4 + 0.4 - 0.04 = 4.36$

ID	Quantity	Target	Error
S623	Rydberg E_R eV	13.6057	EXACT
S624	Stefan σ	5.67	EXACT
S625	Wien b	2.898	0.058%
S626	Faraday F	96485	EXACT
S627	Avogadro N_A	6.022	0.007%
S628	k_B	1.381	0.027%
S629	Planck h	6.626	0.026%
S630	c	2.998	0.033%
S631	e	1.602	0.065%
S632	Hartree E_h	4.36	EXACT

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Highlights. Tier EE delivers **four exact closures** (Rydberg energy 13.6057 eV, Stefan--Boltzmann lead 5.67, Faraday constant 96485 C/mol, Hartree energy lead 4.36) and six near-exact (all under 0.07%). Avogadro closes to 0.007%, Planck and Boltzmann under 0.03%, the speed of light $c = \sqrt{5/D_{\text{phys}} + F_{\text{tr}}/D_{\text{phys}} + F_{\text{tr}}^2/D_{\text{phys}}} = 2.997$ at 0.033%. Cumulative: 337 closures; 91 lockings.